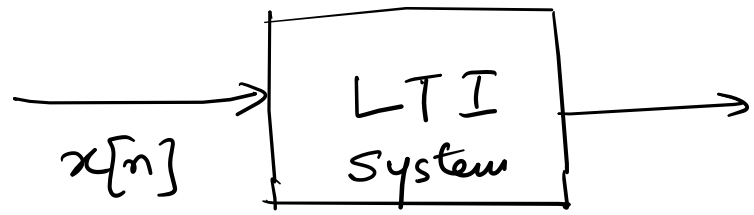
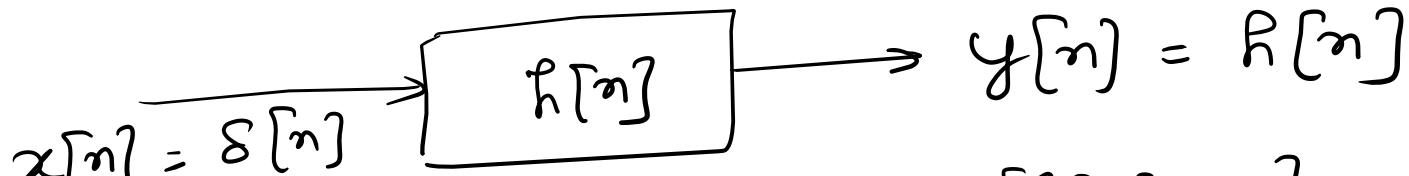
3/9/2025 D.T.

Linear time-invariant Systems



Impulse response of the system is being represented by $h[n]$



$x[1]$ is a constant

$x[1]\delta[n-1]$ → $x[1]h[n-1]$

$x[2]\delta[n-2]$ → $x[2]h[n-2]$

$x[-1]\delta[n+1]$ → $x[-1]h[n+1]$

Sum all these.

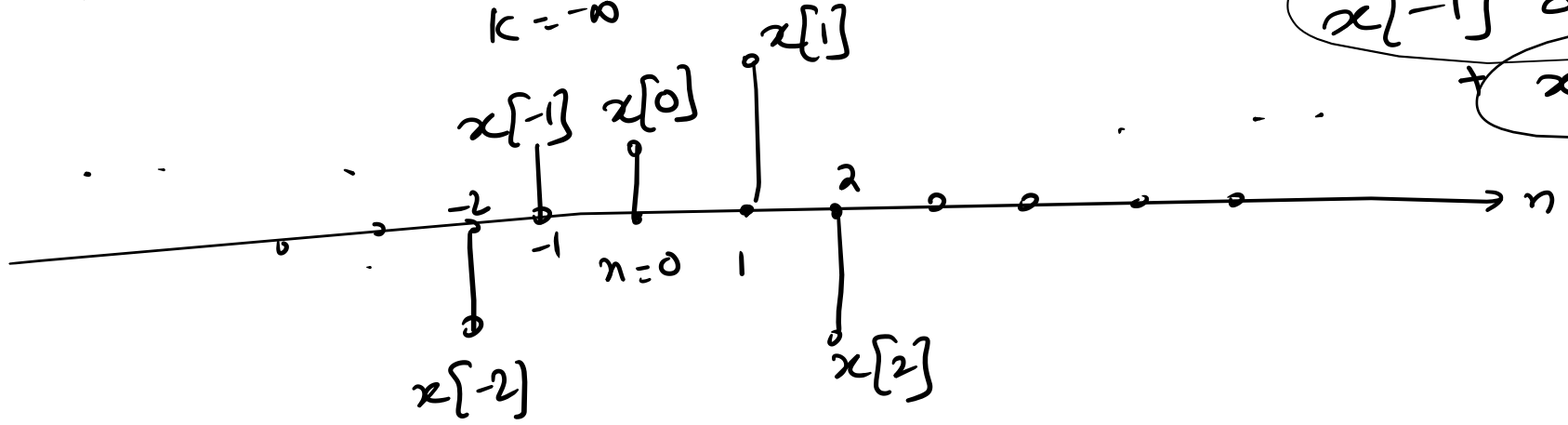
$$\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

i/p =

$$\text{o/p} = y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

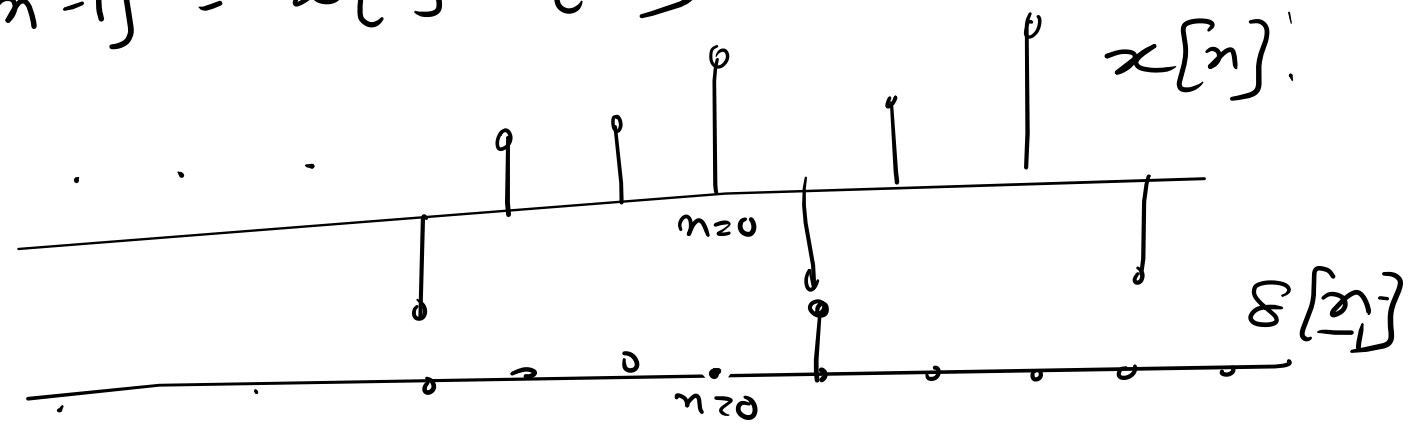
$$x[n] = i/p = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] = \cdots$$

$x[-2]\,\delta[n+2] +$
$x[-1]\,\delta[n+1] +$
$x[0]\,\delta[n] +$
$x[1]\,\delta[n-1] +$
$x[2]\,\delta[n-2] +$
$\cdots$



$$x[n]\,\delta[n] = x[0]\,\delta[n]$$

$$x[n]\,\delta[n-1] = x[1]\,\delta[n-1]$$



$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

$$= x[n] * \delta[n]$$

↑ Symbol for convolution

↓ mathematically

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \longrightarrow \boxed{\text{LTI system}} \longrightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

↓

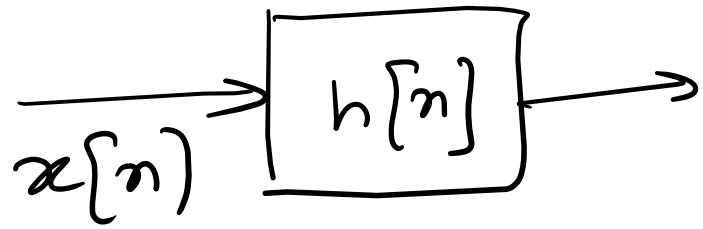
i/p is a weighted linear combination of delayed impulses.

↓

$$y[n] = x[n] * h[n]$$

↑

o/p is a convolution of i/p with impulse response of the system



$$x[n] * h[n] = y[n]$$

$$\left(\alpha_1 x_1[n] + \alpha_2 x_2[n]\right) * h[n]$$

$*$ ↑ operator

$$= \alpha_1 x_1[n] * h[n] + \alpha_2 x_2[n] * h[n]$$

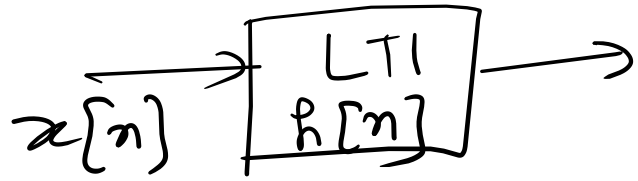
$$x[n] \xrightarrow[\text{system}]{\text{LTI}} y[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$h[n] * x[n]$$

ex



$$h[n] = \begin{cases} 1 \\ 0 \end{cases} \qquad n = 0, 1, 2 \text{ otherwise}$$

$$x[n] = \frac{1}{2}\delta[n] + \frac{2}{3}\delta[n-2] \qquad \frac{2}{3} \times \frac{1}{2}$$

Sol.

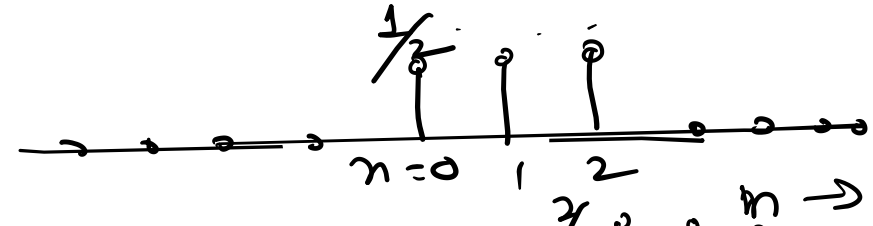
$$h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

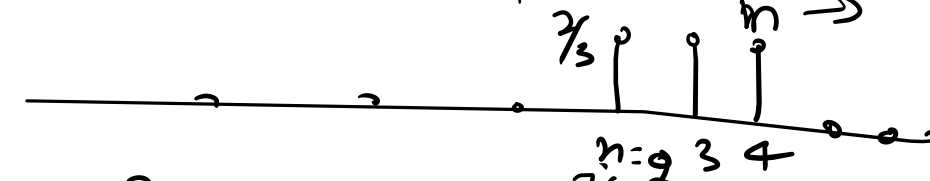
$$\frac{1}{2}\delta[n] \longrightarrow \frac{1}{2}h[n]$$

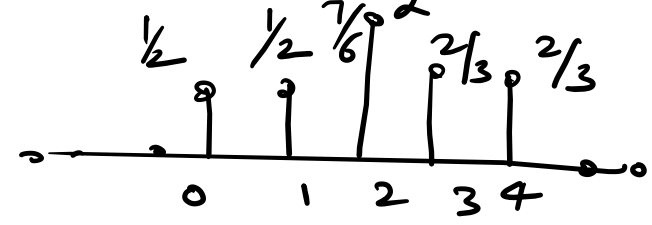
$$+\frac{2}{3}\delta[n-2] \longrightarrow \frac{2}{3}h[n-2]$$

$$x[n] = \frac{1}{2}\delta[n] + \frac{2}{3}\delta[n-2] \longrightarrow \frac{1}{2}h[n] + \frac{2}{3}h[n-2]$$

$\frac{1}{2}h[n]$





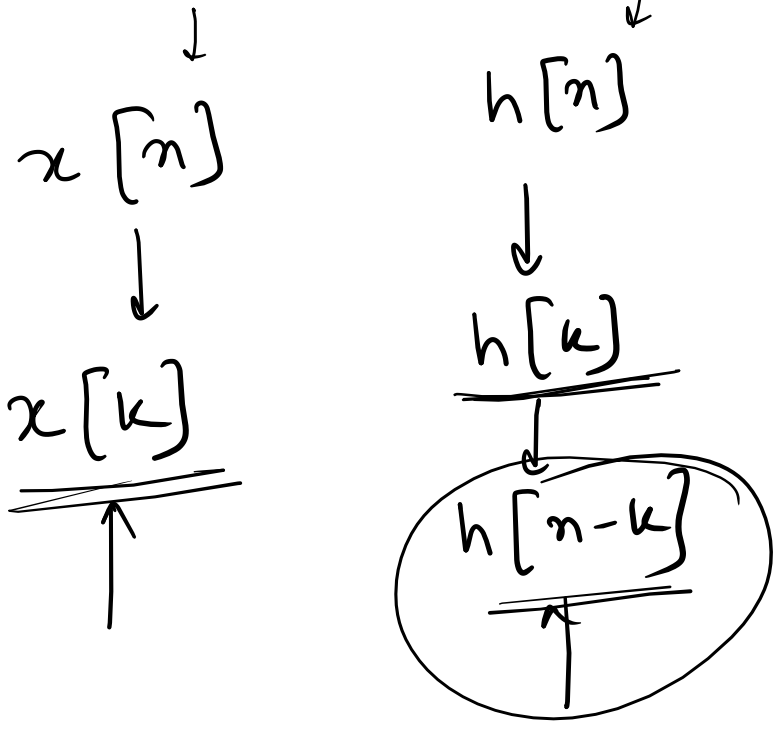


$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

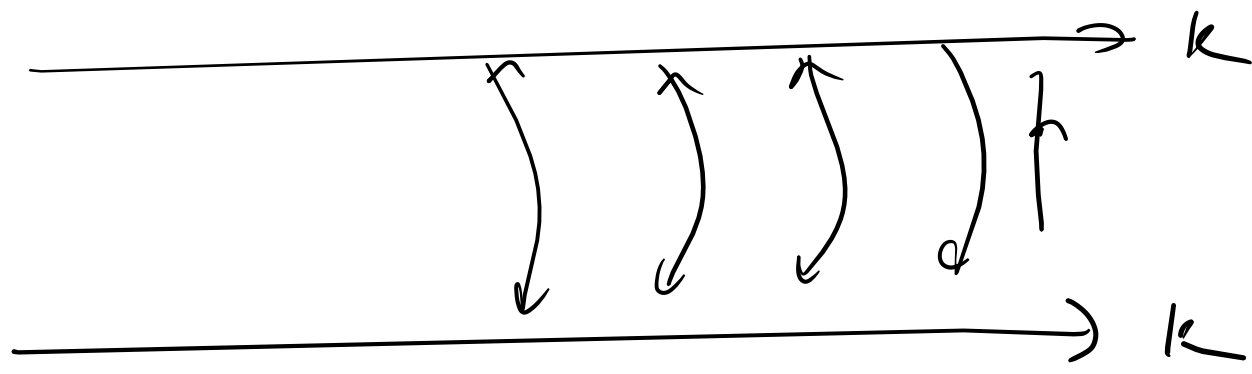
Given $h[n]$ & $x[n]$, compute $y[n]$.

$$x[n] = \alpha^n u[n] \qquad 0 < \alpha < 1$$

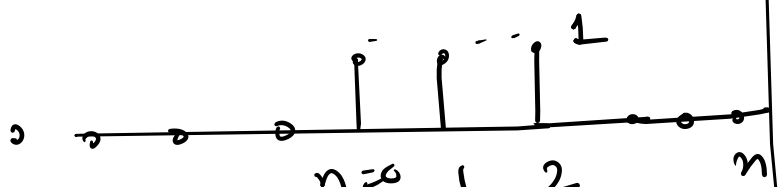
& $h[n] = u[n]$



$x[k]$, $h[n-k]$



Ex: Say $h[n]$



$h[2-k]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]\, h[-k]$$

$$y[1] = \sum_{k=-\infty}^{\infty} x[k]\, h[1-k]$$

$$y[-1] = \sum_{k=-\infty}^{\infty} x[k]\, h[-1-k]$$

$h[-k+2]$ →

$h[-k+n] = h[n-k]$ where n is +ve → shifting $h[-k]$ to the right

when n is -ve → shifting $h[-k]$ to the left.

$x[n]$ → $x[k]$

$h[n]$ → $h[k]$ → $h[-k]$ → $h[n-k]$

$h[-k]$ shifted to right (when n is +ve)

$h[-k]$ shifted to left (when n is -ve)

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]\, h[-k]$$

$y[1] = 1+\alpha$

$y[0] = 1$

$$y[n] = \begin{cases} 0 & n<0 \\ 1+\alpha+\alpha^2 & \\ \sum_{k=0}^{n} \alpha^k & \end{cases}$$

$y[2] = 1+\alpha+\alpha^2$

$y[n] = \sum_{k=0}^{n} \alpha^k$

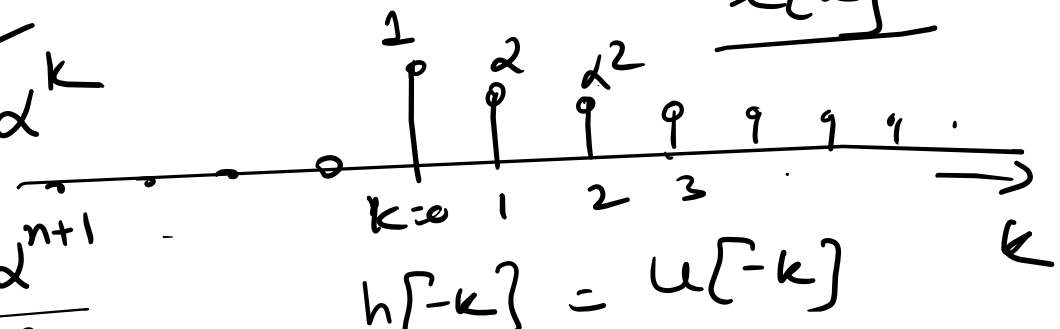
$$= \frac{1-\alpha^{n+1}}{1-\alpha}$$

$0<\alpha<1$

$x[n] = \alpha^n u[n]$

$x[k] = \alpha^k u[k]$

$x[k]$



$h[-k] = u[-k]$

